

RENEWABLE ENERGY RESOURCES AND TECHNOLOGIES IN NIGERIA: PRESENT SITUATION, FUTURE PROSPECTS AND POLICY FRAMEWORK

JOHN-FELIX K. AKINBAMI

Energy/Environment Analysis Group, Centre for Energy Research and Development, Obafemi Awolowo University, Ile-Ife, Nigeria, Tel.: + 234-36-233638 (Office); Fax: + 234-1-4977845; E-mail: jakinbam@energy.cerd.oauife.edu.ng

(Received 18 August 1999; accepted 26 February 2001)

Abstract. Nigeria is endowed with abundant energy resources, both conventional and renewable, which provide her with immense capacity to develop an effective national energy plan. However, introduction of renewable energy resources into the nation's energy mix have implications on its energy budget. The national energy supply system has been projected into the future using MARKAL, a large scale linear optimization model. However, this model may not be absolutely representative of the highly non-linear future of renewable energy. Results of the model reveal that under only a least cost constraint, only large hydro power technology is the prominent commercial renewable energy technology in the electricity supply mix of the country. Despite the immense solar energy potentials available, solar electricity generation is attractive only under severe CO₂ emissions mitigation of the nation's energy supply system. Similarly, the penetration of small-scale hydro power technology in the electricity supply mix is favoured only under CO₂ emissions constraints. Due to economy of scale, large hydro power technology takes the lion share of all the commercial renewable energy resources share for electricity generation under any CO₂ emissions constraint. These analyses reveal that some barriers exist to the development and penetration of renewable energy resources for electricity production in Nigeria's energy supply system. Barriers and possible strategies to overcome them are discussed. Intensive efforts and realistic approach towards energy supply system in the country will have to be adopted in order to adequately exploit renewable energy resources and technologies for economic growth and development.

Keywords: barriers, CO₂ emissions reduction strategy, MARKAL model, mitigation technologies, Nigeria, policy recommendations, renewable energy based electricity, renewable energy resources, renewable energy technologies

1. Introduction

Renewable energy sources involve the harnessing of natural energy flows (e.g., sunlight, wind, waves, falling water, ocean currents, and tides) or the tapping of natural stocks of energy whose rates of replenishment are comparable to or greater than the human use rates (such as ocean thermal gradients, biomass, and hydro-power reservoirs) (Holdren et al. 1980). The World Commission on Environment and Development (WCED) called for renewable energy resources as the foundation of the global energy structure during the 21st century (WCED 1987). This call



Mitigation and Adaptation Strategies for Global Change **6**: 155–181, 2001.

© 2001 Kluwer Academic Publishers. Printed in the Netherlands.

may be based on the theoretical potential estimates of renewable energy resources globally. For instance, the maximum technical and practical potentials of renewable energy resources globally have been estimated to be about 7.04×10^4 and 3.36×10^4 TWh_e per annum respectively by the year 2030 (Swisher et al. 1993). The contribution of the renewable energy resources to total world energy supply is getting more significant. For instance, biomass provides currently about 40–55 Exajouile (EJ) per annum to the world's energy consumption (IIASA/WEC 1998; Larson et al. 1999). At the end of 1998, the installed wind power world wide was about 10,000 MegaWatts (MW) while added capacity was close to 2,600 MW the same year, and another 3,000 MW would have been installed at the end of 1999 (BTM Consult ApS 1999). However, the assessed global wind energy potential that may be realistically harvested is put at 640 EJ per annum (Grubb and Meyer 1993). Several studies have been carried out to assess the potential application of solar energy technologies to world energy consumption through electricity generation to be between 2–16 EJ (WEC 1994; IIASA/WEC 1998; Shell 1996). While total amount of hydropower electricity production world wide reached 2,600,000 GigaWatt-hour (GWh) in 1997 (World Atlas, 1998), electricity generation by and direct use of geothermal energy were 43,756 and 38,178 GWh/annum respectively in 1997 (Stefansson and Fridleifsson 1998).

Nigeria is an energy rich nation with both fossil and renewable fuels. Its renewable energy potential base includes biomass (animal, agricultural and wood residues, fuelwood), solar, hydro, wind and geothermal. Only biomass and hydropower have been and are still being exploited. In recent times, the rate of exploitation of fuelwood is greater than its regeneration. If adequate intervening measures are not quickly implemented fuelwood may no longer be thought of as renewable in the country. In Nigeria, the prejudice for fossil fuels to the neglect of the abundant renewable energy resources is often acknowledged as improper. Shehu Shagari, the former President of the Federal Republic of Nigeria put this clarion call more succinctly in his message to the 'U.N. Nairobi Conference on New and Renewable Sources of Energy' in 1981 when he said: '...For us developing countries which are moving from energy patterns dominated by wood fuels to more commercial energy forms, we now have an opportunity to take action before stumbling into a fossil fuel trap. We ought now to investigate the possibilities in at least three directions (U.N. 1981):

- We need to reinvestigate the use of crop and wood plantations as basis of renewable energy supplies. Fortunately, much of the developing countries lie in equatorial and tropical zones where such development is eminently feasible
- Solar energy seems an appropriate form given the very size of that resource. Here, considerations of technology are of course paramount
- In an energy scarce period, we cannot afford to leave unutilized the potential for hydro-electric power schemes. Opportunities that remain in this field are in developing countries'.

TABLE I
Bioenergy potential of Nigeria (1990) (Gaafar and Kouakou 1993)

Biomass resource	Animal residue	Agricultural residue	Wood residue (Industrial Wood, Fuelwood, Charcoal)	Total
Potential (GJ)	47,718	325,822	805,580	1,179,120

Many national policy decisions taken today could well affect future climate change prospects significantly. Hence an effective and practical climate change strategy that is more convincing to decisionmakers, as well as integrate and reconcile the development, equity, and sustainability (DES) aspects within a holistic and balanced sustainable development framework, has to be developed (Munasinghe 1999). This paper presents suggestions to a practical climate change and sustainable development strategy by putting the renewable energy resources and technologies in perspective in Nigeria.

2. Estimates of Potential Renewable Energy Resources

A modest estimate of the technical potential of solar energy in Nigeria with 5% device conversion efficiency is put at 15.0×10^{14} KiloJoule (KJ) of useful energy annually (Onyebuchi 1989). This is about 258.62 million barrels of oil equivalent annually which compares with current national annual fossil fuel production in the country. This will also amount to about 4.2×10^5 GWh of electricity production annually, which is about 26 times the recent annual electricity production of 16,000 GWh in the country. In 1990, energy content of total national biomass potential stood at approximately 1.2 PetaJoule (PJ) (Table I). Currently, total hydro electric power potential is estimated to be about 8,824 MW with an annual electricity generation potential in excess of 36,000 GWh. This consists of 8,000 MW of large hydro power technology while the remaining 824 MW is small-scale hydro power technology. Presently, 24% and 4% of both large and small hydro power potentials, respectively, in the country have been exploited. Various studies including Lovejoy (1984), Ojosu and Salawu (1990) reveal that the potential for wind energy abound in northern Nigeria. The potential estimates of geothermal energy are yet to be carried out. There are considerable potentials for ocean thermal energy conversion (OTEC) and biogas production in Nigeria.

TABLE II
Distribution of energy consumption in Nigeria (Calculated from FOS, 1980, 1985, 1990, 1995)

	1980	1985	1990	1995
Coal (PJ)	2.75	3.86	2.14	0.57
Natural gas (PJ)	2.20	45.73	32.78	44.99
Hydro (PJ)	9.91	10.83	10.09	26.35
Fuelwood + charcoal + other biomass (PJ)	340.77	385.21	412.29	464.85
Petroleum products (PJ)	297.94	388.83	414.28	355.95
Electricity (PJ)	18.50	26.44	28.33	35.35
Total energy consumption (PJ)	653.57	834.46	871.58	901.71
Total commercial energy (PJ)	312.80	449.25	459.29	436.86
Fuelwood + charcoal + other biomass + hydro (PJ)	350.68	396.04	422.38	491.20
Share of electricity in total commercial energy (%)	5.91	5.89	6.17	8.09
Share of electricity in total energy (%)	2.83	3.17	3.25	3.92
Share of petroleum products in total commercial energy (%)	95.25	86.55	90.20	81.48
Share of petroleum products in total energy (%)	45.59	46.60	47.53	39.47
Share of hydro in commercial energy (%)	3.17	2.41	2.20	6.03
Share of hydro in total energy (%)	1.52	1.30	1.16	2.92
Share of total biomass in total energy (%)	52.14	46.16	47.30	51.55
Share of biomass + hydro in total energy (%)	53.66	47.46	48.46	54.47

TABLE III
Share of fuel use for electricity generation (%) (Calculated from FOS, 1980, 1985, 1990, 1995)

Year	Natural gas	Coal	Diesel	Fuel oil	Hydro	Total fossil fuel	Total renewable energy
1980	77.54	1.07	2.99	1.14	17.26	82.74	17.26
1985	82.89	0.15	5.31	0.77	10.88	89.12	10.88
1990	88.15	0.14	0.25	0.67	10.79	89.21	10.79
1995	73.51	0.00	0.02	0.64	25.83	74.17	25.83

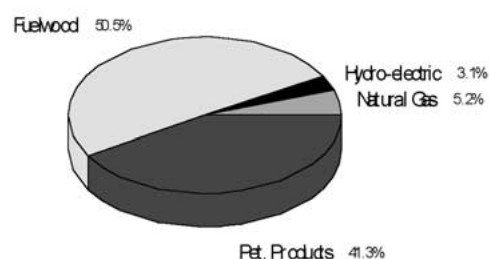


Figure 1. Typical current energy supply mix in Nigeria (1995) (FOS 1996).

3. Current Status of Renewable Energy Use

Table II shows that there has been a steady increase in the consumption patterns of both hydro and biomass resources in Nigeria. However, the share of hydro in the commercial and total energy consumption is very low annually. While the share of hydro in commercial energy consumption ranged between 3–6%, its share in total energy consumption was in the range 1–3% during 1980–1995. On the other hand, the share of biomass in total energy consumption averaged about 50% in the same period and the share of both biomass and hydro together in the total energy consumption followed the same trend. Commercial renewable energy is presently of immense benefit to electricity generation in the country. In the period 1970–1990, the generation mix was dominated by hydro and natural gas thermal plants. Between 1973 and 1978, hydro electric power took the largest share, about 46% of the total installed capacity (IER/CERD 1990). From the early 1980s, the balance has shifted in favour of gas thermal plants as depicted by the share of fuel use for electricity generation in Table III. The 1995 distribution of energy consumption, as revealed by Figure 1, typifies the current energy supply mix in the country. This pie chart shows that of the total energy consumption, the share of natural gas was 5.22%, hydroelectricity took 3.05%, fuelwood had the lion share of 50.45% and petroleum products had 41.28% share. This further confirms the fact that presently, renewable-energy use in the country is split essentially between hydroelectricity and traditional fuelwood. Therefore, commercial renewable energy resources contribute only a small share of total commercial electricity supply in the current national energy supply system.

4. Future Prospects of Renewable Energy Use

4.1. METHODOLOGY

In examining the future prospects of renewable energy in the national economy, the primary tool employed for the analysis is a multi-period linear programming optimization model called MARKAL (**MARK**et **AL**location) (Fishbone et al. 1983).

TABLE IV

Key assumptions regarding the parameters used in the MARKAL model for this study

1	Socio-economic
(a)	Demographic
(i)	LEG Scenario – Population growth rate is assumed at the constant rate of 2.8%/annum throughout the study period with the understanding that due to poor economic conditions, no attempt is made at family planning
(ii)	MEG Scenario – Population growth rate decreases from 2.8%/annum to about 2.1%/annum at the end of the study period. This is with the understanding that some moderate measures at family planning to reduce population are adopted.
(iii)	HEG Scenario – Population growth rate is assumed to decrease from 2.8%/annum to about 1.4%/annum at the end of the study period with the hope that some aggressive measures to reduce population are adopted.
(b)	Economic
(i)	LEG Scenario – GDP follows a logistic growth pattern with 3.0%/annum over the study period. Incomes are also assumed to follow the GDP pattern
(ii)	MEG Scenario – GDP follows a linear growth pattern with 5.0%/annum over the study period, incomes similarly follow the GDP pattern.
(iii)	HEG Scenario – GDP follows an exponential pattern with a growth rate of 6.0%/annum over the study period and incomes follow similar trend.
2.	Prices of oil and gas
(i)	Prices of Crude Oil: There were large variations in oil prices in 1990 due to the Gulf war. These ranged from \$18.92/barrel to \$36.95/barrel and again down to \$28.14/barrel. We assumed an export cost of \$19.69/barrel for 1990, thereafter coming down to the post-Gulf crisis value of \$17/barrel in 1995. The projected costs for 2010 and 2030 are \$24 and \$28 per barrel respectively. These are as suggested in the UNEP Guidelines (1994) for the global reference scenario.
(ii)	Prices of Natural Gas: When all the projects under way for exporting gas are completed, it is expected that by the year 2001, according to a World Bank Report (1993), gas export price will be \$27.84 per 1,000 m ³ .

This model investigates the details of energy supply systems by evaluating energy technologies given the criterion to be optimised. The versions of the MARKAL model are widely used in the world as nation-wide energy models (Fujii and Yamaji 1998). The model was used to study the energy supply system of Nigeria for the period 1990–2030 with the objective function of maximizing the supply system at the least cost to the system. Although MARKAL is not a predictive model, it can identify the least cost path within the reference energy system (RES) that best satisfies the overall objective of the energy-environmental system. In this respect, useful energy demand projections were computed using a simulation model called MADE-II (Model for Analysis of Demand for Energy). MADE-II employs a

TABLE V
Technology characteristics of electricity conversion technologies used in this study

Technology	Capital cost \$/KW	Variable operating and maintenance cost \$/KW/yr.	Fixed operating and maintenance cost \$/KW/yr.	Technical efficiency	Capacity utilization factor	Sources of data
Central solar voltaic	2638.89	12.70	10.80	0.3173	0.84	EPRI TAG (1993) Technology # 22.1C
Residential solar PV	12240– 3825	12.70	10.80	0.3320– 0.3500	0.84	Annual energy outlook, 1991, Table C6
Small hydro	2107	–	11.99	0.3128	0.85	EPA's inventory of technologies program, 1995, p. 74
Mini hydro	2632	–	14.08	0.3128	0.85	EPA's inventory of technologies program, 1995, p. 74
Micro hydro	3156	–	16.16	0.3128	0.85	EPA's inventory of technologies program, 1995, p. 74
Zungeru large hydro	1800	–	27.00	0.3128	0.85	World Bank ESMAP Report, 1993a, p. 97
Mambilla large hydro	1800	–	27.00	0.3128	0.85	World Bank ESMAP Report, 1993a, p. 99
Guregu gas turbine	400	1.79	22.00	0.2801	0.85	Beicip, 1991, p. 30
Jebba hydro	1800	–	27.00	0.3128	0.340– 0.387	World Bank ESMAP Report, 1983 p. 100
Kainji hydro	1800	–	27.00	0.3128	0.1722	World Bank ESMAP Report, 1983 p. 100
Shiroro hydro	1800	–	27.00	0.3128	0.80	World Bank ESMAP Report, 1983 p. 100
Delta gas turbine	400	1.79	22.00	0.2283– 0.2439	0.30	World Bank ESMAP Report, 1993a
Sapele gas turbine	400	1.79	22.00	0.2315	0.80	Beicip, 1991, p. 8 and World Bank ESMAP Report, 1993a
Sapele steam turbine	850	1.79	29.75	0.2857– 0.3413	0.80	Beicip, 1991, p.8, 30 and World Bank ESMAP Report, 1993a
Afam gas turbine	400	1.79	22.00	0.2132	0.80	Beicip, 1991, p. 8 and World Bank ESMAP Report, 1993a
Egbin steam turbine	850	1.79	29.75	0.40	0.40	Beicip, 1991, p. 8, 30 and World Bank ESMAP Report, 1993a
Combined cycle (new)	650	1.79	29.25	0.46	0.85	Beicip 1991, p. 30, 35
Gas turbine (new)	400	1.79	22.00	0.2801	0.85	Beicip 1991, p. 30, 35
Conventional gas thermal (new)	850	1.79	29.75	0.36	0.85	Beicip 1991, p. 26, 30, 35
Conventional steam turbine (new)	900	1.00	31.50	0.36	0.85	Beicip 1991, p. 26, 30, 35

Costs are in US\$ 1990.

TABLE VI

Installable capacity and capacity addition of renewable energy technologies used in the MARKAL model

Renewable energy technology considered	Installable capacity	Annual capacity addition at year of entry into energy supply system
Small hydro power plant (1.0–10.0 MW unit installed capacity)	704.1 MW	20% per annum of the installable capacity
Micro hydro power plant (< 0.5 MW unit installed capacity)	24.5 MW	10%% per annum of the installable capacity
Mini hydro power plant (0.5–1.0 MW unit installed capacity)	95.6 MW	10%% per annum of the installable capacity
Residential solar photo-voltaic	100 MW	10% per annum of the installable capacity
Central solar photo-voltaic	500 MW	Gradually built into the system to reach the maximum installable capacity

combination of statistical, econometric and engineering process techniques in calculating useful energy demand projections (Saboohi 1989). Five economic sectors – residential, agricultural, commercial, industrial, and transportation were considered. In earlier works (Ibitoye et al. 1998a, 1998b; Ibitoye and Akinbami 1999), the MARKAL model was used to study the cost of carbon-dioxide (CO₂) abatement in Nigeria's energy sector, the optimal structure of Nigeria's energy sector under CO₂ reductions and to make a case for strategies to implement CO₂ mitigation options in Nigeria's energy sector. Tables IV and V summarize the key input assumptions on parameters used in the models for this study.

4.2. SCENARIOS ADOPTED

Due to the uncertain evolution of developing country economies which the Nigerian economy typifies, we considered three energy supply scenarios which assume different patterns of energy development. These are the LEG scenario which represents slow economic growth and hence low energy growth, the MEG scenario which assumes moderate economic growth and development and hence, a moderate energy growth and HEG, the high energy growth, which represents strong economic and social growth. It is expected that Nigeria's energy growth pattern will be somewhere between the LEG and HEG scenarios. In order to examine the Nigerian energy supply system under CO₂ mitigation strategies as part of efforts to address the climate change issues identified in the United Nations Framework Convention on Climate Change (UNFCCC) which Nigeria has already ratified, two CO₂ emission reduction levels were imposed on the energy system. These

TABLE VII
Default CO₂ emission factors (IPCC 1995)

Fuel type	Emission factor (kg CO ₂ /GJ)
Coal	101.2
Oil	73.3
Natural Gas	56.1
Gasoline	69.3
Diesel	74.1
Fuel Oil	77.4
Kerosene	71.3
Liquefied petroleum gas	63.1
Condensate	55.7

strategy scenarios are 20% and 35% reduction by 2030 (from the 2030 level of the baseline) assuming in each case a linear reduction starting from 2000. A 20% CO₂ mitigation in this study was carried out by reducing the CO₂ level relative to the baseline emission curve starting from 2000 and making it reach the desired 20% level by 2030. The mitigation exercise was carried out on all the three scenarios. This implies that each scenario LEG, MEG and HEG has three sets of emission levels consisting of the baseline, 20% and 35% CO₂ reduction levels. The baseline is the non-CO₂ constrained result for the three scenarios and it follows the concept of the most likely development in which energy projects that are being planned or proposed are assumed to be implemented. The current inefficiencies in the energy system are not necessarily carried forward into the future as is the case in a business-as-usual approach. Energy technology options introduced into the system for the mitigation of CO₂ include demand-side management, increased gas-use, supply-side management, and increased use of renewable energy technologies. The increased use of renewable energy technologies option is reflected in the use of small, mini, and micro-hydro power plants, central photovoltaic, and residential solar photovoltaic (PV). Table VI provides an overview of the annual capacity addition for each of these technologies as entered into the MARKAL model. The technologies are built up into the national energy supply system to reach the maximum installable capacity by the end of the study period. Two new large hydroelectric plants are considered in the baseline within the study period and they sum up to 3250 MW installed capacity. Results of the optimisation and mitigation analyses carried out in the Nigeria national energy supply system are presented in subsequent sub-sections.

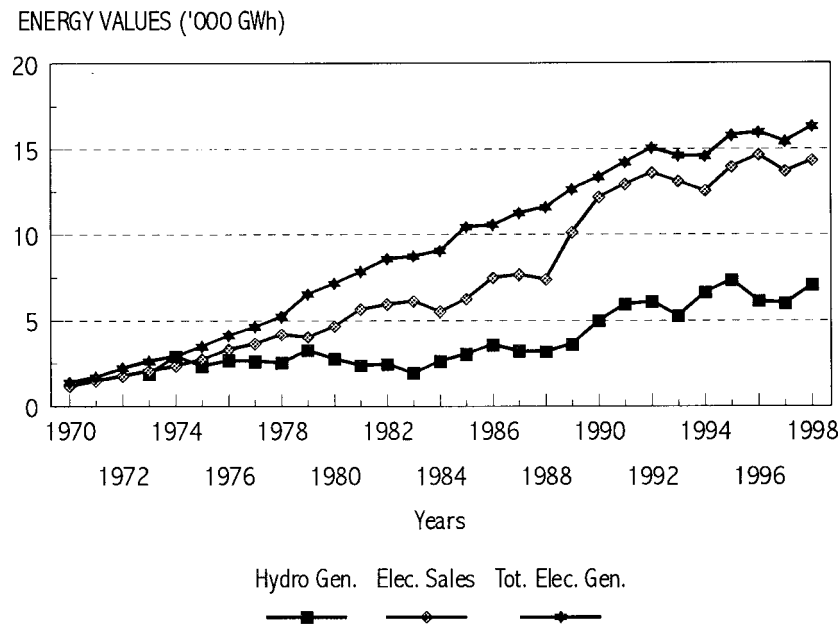


Figure 2. Electricity generation and consumption trends 1970–1998. These data do not take into account private generation of electricity. Sources: (1) Annual abstract of statistics, (FOS), Nigeria. (2) Annual technical report – NEPA generation and transmission grid operations, National Electric Power Authority (NEPA) National control centre, Osogbo, Nigeria. Hydro Gen. (■) is total hydro-power generated; Elec. Sales (◈) is total electricity sold or consumed; Tot. Elec. Gen. (★) is total

electricity generated in the country.

4.3. CO₂ EMISSION COEFFICIENTS

Greenhouse gases (GHGs) include CO₂, methane (CH₄), nitrogen oxide (N₂O), and chloro Fluorocarbons (CFCs). Some of these GHGs have a greater effect on global warming than others. Carbon dioxide is responsible for more than half of the total radiative forcing because of its large absolute increase in atmospheric concentration (Fujii and Yamaji 1998). This is the basis for the CO₂ mitigation analysis in this study. In estimating the emission coefficients, the Intergovernmental Panel on Climate Change (IPCC) Reference Approach on emissions accounting has been adopted. This is based on accounting for the C in fuels supplied to the economy, irrespective of technologies consuming the fuel or whatever transformations the fuel went through before being finally consumed. Since appropriate national emission factors are not available yet for Nigeria, the IPCC default CO₂ emission factors have been adopted in this study. Table VII gives the values of the emission factors of the various fuels used.

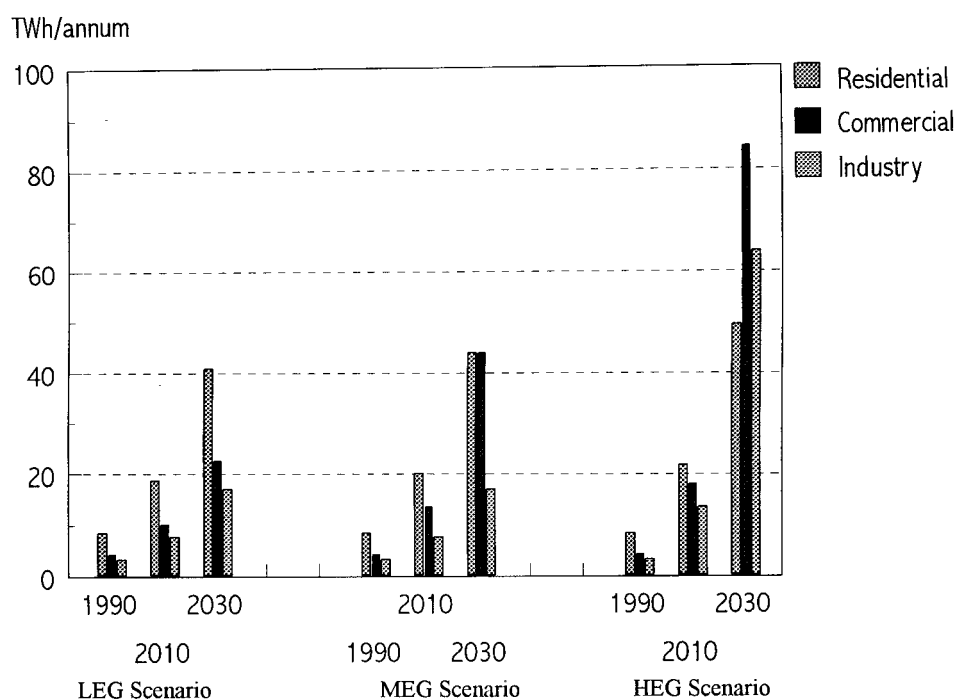


Figure 3. Sectoral electricity demand projections in TeraWatt-hours (TWh).

5. Results and Discussion

5.1. CURRENT AND FUTURE ELECTRICITY GENERATION AND USE

Nigeria's electric power supply system is an interconnected national grid system. Installed capacity has grown from 700 MW in 1970 to about 6,000 MW at the end of 1990. The available capacity (about 2,200 MW) is just about one-third of the total installed capacity. This is due to several operational handicaps which have impaired capacity. Generation has grown from 1,500 GWh in 1970 to over 16,000 GWh in 1998 while consumption has risen from 1,147 GWh in 1970 to about 14,325 GWh in 1998. Electricity demand is mainly from residential, commercial, and industrial sectors of the nation's economy. While electricity consumption per capita was low from 1970–1990, it grew much faster than the GDP per capita revealing that Nigeria is developing rapidly. In 1992, rural consumers accounted for more than 70% of the population, but less than 20% of rural dwellers had access to electricity. However, 82% of the urban dwellers had access to electricity in the same year (FOS 1992). Electricity demand will continue to increase since it is necessary to maintain modern life and is an essential input to economic growth and development. Electricity consumption in the country grossly underestimates actual demand because of suppressed demand which often results from

load shedding. Load growth has been constrained by the inability of the network to accommodate new customers, power unreliability, and shortage of available capacity. Consequently, industrial firms, affluent households and commercial enterprises have had to depend on private diesel and petrol fueled electric generating plants. While Figure 2 shows electricity consumption for 1970–1998, Figure 3 depicts the sectoral electricity demand projections for the country from the energy demand model used – MADE-II.

Total electricity production in 1990 was 14.92 TeraWatt-hour (TWh) and is projected to increase 4-fold to 52.07 TWh in 2030 in the LEG scenario. Similarly, under non-CO₂ constraint strategy, electricity production in 2030 increases about 5- and 8-fold over the base year production value in the MEG and HEG scenarios, respectively. Under CO₂ constraints, the electricity production profile is different. For instance, under the 20% CO₂ reduction strategy, total electricity production increased from 14.92 TWh in 1990 to 49.97 TWh in 2030 in the LEG scenario. Similarly, the base year value increased to 68.08 and 106.64 TWh in 2030 for the MEG and HEG scenarios respectively. Similar trends are observed for the 35% CO₂ reduction strategy. The observed fall in electricity production levels in the CO₂ reduction strategies for all the scenarios is probably associated with the increased use of decentralized electric conversion technologies and improved energy efficiency. In 1990, the shares of these sectors to the total electricity consumption were about 21%, 26% commercial industrial and 53% for the commercial, industrial and residential sector, respectively. By 2030, the residential sector consumes about 49% while industry and commercial sectors take 27% and 24% respectively in the LEG scenario. Similar trends are observed for the 20% and 35% CO₂ emission reduction strategies for the LEG scenario. Under the baseline strategy, while industry share in total electricity consumption is 36% in 2030, residential and commercial sectors have 33% and 31% respectively in the MEG scenario. In the HEG scenario, the industry sector has the highest total electricity consumption followed by commercial and residential sectors, respectively, under all CO₂ emission reduction strategies.

5.2. RENEWABLE ENERGY IN ENERGY CONSUMPTION TREND

Analysis reveals that the Nigerian energy supply system is fossil fuel dominated under all economic, energy growth and under CO₂ emissions mitigation scenarios. Tables VIII, IX and X reveal a steady increase in fossil fuel (coal, natural gas, crude oil) consumption under all scenarios and CO₂ reduction strategies. A steady decline in consumption is observed for renewable energy (including biomass) under the MEG and HEG scenarios. These scenarios represent better economic growth and development, and a transition from inefficient end-use technologies and traditional fuels to more efficient technologies. However, the opposite trend is observed under the LEG scenario. This is because of slow economic growth and development which results in low disposable incomes. Consequently, a greater percentage

TABLE VIII

Energy consumption trends in Nigeria's economy for the baseline (non-CO₂ emissions reduction strategy) (PJ)

	LEG scenario			MEG scenario			HEG scenario		
	1990	2010	2030	1990	2010	2030	1990	2010	2030
Fossil fuel	1988.80	2651.60	3768.50	1988.80	2692.00	4083.10	1988.80	2915.80	5249.90
All									
renewable									
energy	557.65	631.33	817.30	557.65	311.65	286.20	557.65	385.35	256.50
Renewable									
energy									
(non-biomass)	9.65	50.23	117.80	9.65	50.95	117.80	9.65	50.95	117.80

TABLE IX

Energy consumption trends in Nigeria's economy for the 20%CO₂ emissions reduction strategy (PJ)

	LEG scenario			MEG scenario			HEG scenario		
	1990	2010	2030	1990	2010	2030	1990	2010	2030
Fossil fuel	1988.80	2611.30	3026.50	1988.80	2614.40	3199.20	1988.80	2804.90	4095.30
All									
renewable									
energy	557.65	609.31	783.60	557.65	365.85	317.55	557.65	376.63	274.19
Renewable									
energy									
(non-biomass)	9.65	48.71	135.90	9.65	58.55	149.15	9.65	58.83	162.39

of the population will rely on biomass and inefficient end-use cooking devices. While fossil fuel consumption is high in absolute terms, commercial renewable energy penetrates the Nigerian energy market at very high rates. Under the MEG scenario, the increase in consumption from 1990 to 2030, is between 12- and 16-fold. Furthermore, the demand for commercial renewable energy increases with increase in the CO₂ emissions constraint.

5.3. GROWTH TREND OF COMMERCIAL RENEWABLE ENERGY

In Nigeria the main application of non-biomass or commercial renewable energy is for electricity generation. While total electricity production was 14.92 TWh in

TABLE X

Energy consumption trends in Nigeria's economy for the 35% CO₂ emissions reduction strategy (PJ)

	LEG scenario			MEG scenario			HEG scenario		
	1990	2010	2030	1990	2010	2030	1990	2010	2030
Fossil fuel	1988.80	2440.60	2641.60	1988.80	2343.10	2747.60	1988.80	2702.10	3390.40
All renewable energy	557.65	617.04	784.92	557.65	318.82	398.90	557.65	377.14	287.44
Renewable energy (non-biomass)	9.65	56.44	137.22	9.65	50.02	151.80	9.65	59.34	175.64

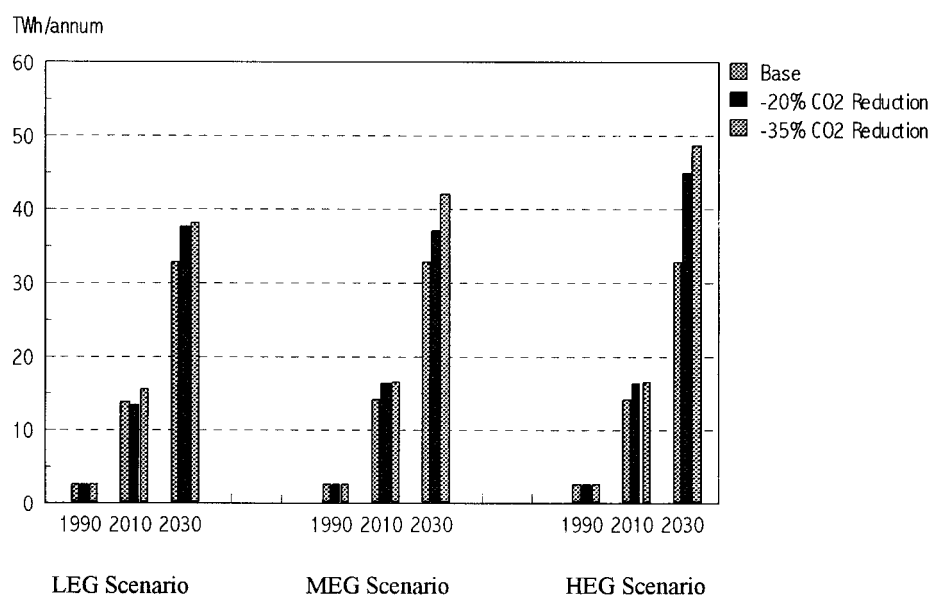


Figure 4. Future trends of total renewable electricity production under different scenario assumptions.

1990, renewable energy contributed 18% while the remainder went to natural gas and oil based thermal plants. Under the baseline (non-CO₂ emission constraint) strategy, the share of renewable energy in electricity production increased to 47%, 45% and 38% of total production in the LEG, MEG, and HEG scenarios, respectively. Similar trends are observed for the year 2030 for the three scenarios. Figure 4 presents the likely trend of renewable energy electricity production profiles in the country for all the scenarios. The analysis reveals that under non-CO₂ emissions

TABLE XI

Growth trends of individual non-biomass renewable energy in national electricity supply mix for the baseline strategy (TWh)

	LEG scenario			MEG scenario			HEG scenario		
	1990	2010	2030	1990	2010	2030	1990	2010	2030
Large hydro	2.68	13.95	32.72	2.68	14.15	32.72	2.68	14.15	32.72
Small hydro	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Central									
solar PV	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Residential									
solar PV	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00

Note: Small hydro is a generic name for the combination of small-, micro-, and mini-hydro power plants. Their definitions still remain as already pointed out in the text.

TABLE XII

Growth trend of individual non-biomass renewable energy in national electricity supply mix for the 20% and 35% CO₂ emissions reduction strategies (TWh)

	20% CO ₂ emissions reduction strategy								
	LEG scenario			MEG scenario			HEG scenario		
	1990	2010	2030	1990	2010	2030	1990	2010	2030
Large hydro	2.68	11.38	32.72	2.68	14.11	28.44	2.68	14.15	32.72
Small hydro	0.00	2.15	5.03	0.00	2.15	5.03	0.00	2.15	5.03
Central									
solar PV	0.00	0.00	0.00	0.00	0.004	3.68	0.00	0.004	7.36
Residential									
solar PV	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	35% CO ₂ emissions reduction strategy								
	LEG scenario			MEG scenario			HEG scenario		
	1990	2010	2030	1990	2010	2030	1990	2010	2030
Large hydro	2.68	13.52	32.72	2.68	14.15	32.72	2.68	14.15	32.72
Small hydro	0.00	2.15	5.03	0.00	2.15	5.03	0.00	2.15	5.03
Central									
solar PV	0.00	0.00	0.00	0.00	0.37	3.68	0.00	0.004	7.36
Residential									
solar PV	0.00	0.00	0.37	0.00	0.00	0.07	0.00	0.01	3.68

TABLE XIII

Distribution of centralized and decentralized renewable energy in total non-biomass renewable energy production (TWh)

	LEG scenario			MEG scenario			HEG scenario		
	1990	2010	2030	1990	2010	2030	1990	2010	2030
Emission levels									
BASELINE									
Centralized	2.68	13.95	32.72	0.00	0.00	0.00	2.68	14.15	32.72
Decentralized	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
20% CO ₂ REDUCTION									
Centralized	2.68	11.38	32.72	2.68	14.11	32.12	2.68	14.19	40.07
Decentralized	0.00	2.15	5.03	0.00	2.15	5.03	0.00	2.15	5.03
35% CO ₂ REDUCTION									
Centralized	2.68	13.52	32.72	2.68	14.52	36.40	2.68	14.19	40.07
Decentralized	0.00	2.15	5.39	0.00	2.15	5.76	0.00	2.29	8.71

constraint strategy and for all the scenarios only large hydro enters into the energy supply system as renewable energy for electricity production (Table XI). This is most likely due to the cost since the objective function of MARKAL model for this study is energy supply optimization at a least cost to the economy. Table V gives the cost parameters of the various electricity conversion technologies used in the model and large hydro has the least costs particularly capital and variable operating and maintenance costs. Other renewable conversion technologies for electricity production (small scale hydro power, central solar PV system, residential solar PV system) enter into the energy supply system only under CO₂ emissions constraint strategies. Of these, only small scale hydro power plants play a significant role in all the scenarios while central solar PV and residential solar PV technologies make significant entry only under MEG and HEG scenarios in 2030 (Table XII). This further underscores the fact that while the decentralized plants are highly probable options for CO₂ mitigation their high investment costs place them at a disadvantage to the large hydro power plants.

5.4. DISTRIBUTION OF RENEWABLE ENERGY BASED ELECTRICITY PRODUCTION

Electricity generation in Nigeria is achieved through centralized and decentralized systems. While the centralized system is connected to the national grid for transmission and distribution to the consumers, the decentralized is usually a stand-alone system which is not connected to the national grid. The centralized system consists of both thermal plants (natural gas, fuel oil and diesel) as well as renew-

able energy based (large hydro power and central PV) plants. The decentralized system similarly consists of both renewable energy based and thermal plants. The renewable energy based decentralized plants are small (small, mini, and micro) hydro power plants and residential PV systems. Table XIII reveals that under the non-CO₂ reduction strategy, total renewable energy based electricity production is from only the centralized system and it increases from 2.68TWh in 1990 to 32.71 TWh in 2030 for all the scenarios. However, under the 20% CO₂ emissions reduction strategy, both centralized and decentralized systems jointly contribute to total renewable energy based electricity. While the centralized system consists of both large hydro and central solar PV plants, only small hydro power plants contribute to the decentralized system. The comparative contribution of centralized to decentralized electricity is in excess of 5-fold in the LEG scenario while under both MEG and HEG scenarios, it is in excess of 6-fold. For the MEG scenario, contribution of the centralized system in 2030 is in excess of 13-fold from the base year value of 2.68 TWh. Under the HEG scenario, the centralized system contribution rose from 2.68 TWh in 1990 to 40.06 TWh in 2030. Central solar PV technology provides about a 19% share while the rest is achieved via large hydro power plants. Similarly, the decentralized system increased from 2.29 TWh in 2010 to 8.70 TWh in 2030. This implies that within the study period time of 40 years, decentralized electric power system would increase just about four-fold. Renewable energy technologies are highly probable options for CO₂ mitigation in Nigeria's energy system but their exorbitant investment costs place them at a disadvantage to the large hydro power plants and natural gas combined cycle plants.

5.5. CARBON-DIOXIDE EMISSIONS TREND

Total CO₂ emissions from the energy sector in 1990 was 90 million Mg. This is expected to increase at an annual rate of 2.0%, 2.6%, and 3.2% in 2030 for the baseline or non-CO₂ emission reduction strategy in the LEG, MEG and HEG scenarios, respectively. Main contributions to the total base year emissions were 38% from gas flaring in the oil sector, 15% each from both the transport and electricity sectors, 8% from the residential sector and 6% from the industrial sector. Under the baseline strategy, a decline in the contribution to total emissions due to the residential sector is observed across the three scenarios. This trend can be attributed to the declining population growth rate in the MEG and HEG scenarios and the economic growth assumptions in the model which improve from the LEG to HEG scenario. A decline in emissions due to gas flaring is observed because more gas which would have been flared is being consumed economically in the HEG scenario in power generation and industrial process heat. In this study, the representation of gas flaring is such that the model, during optimization, could choose to continue to flare associated gas at the present rate, or the model could decide to recover more of the associated gas destined for flaring. However, since the cost of associated gas recovery is higher than that of non-associated gas production, it is not likely that

TABLE XIV
CO₂ emissions trends (million Mg) in Nigeria.

Scenario	LEG scenario		MEG scenario		HEG scenario	
	1990	2030	1990	2030	1990	2030
BASELINE	90	199	90	254	90	329
20% CO ₂ REDUCTION	90	159	90	203	90	261
35% CO ₂ REDUCTION	90	126	90	162	90	214

TABLE XV
CO₂ emissions from major sectors (million Mg) in Nigeria.

	20% CO ₂ emissions reduction						35% CO ₂ emissions reduction					
	LEG		MEG		HEG		LEG		MEG		HEG	
	scenario		scenario		scenario		scenario		scenario		scenario	
	1990	2030	1990	2030	1990	2030	1990	2030	1990	2030	1990	2030
Electricity												
generation	13.50	14.66	13.50	23.40	13.50	37.44	13.50	10.48	13.50	7.02	13.50	25.00
Gas flaring	34.20	16.71	34.50	9.38	34.20	3.91	34.20	2.38	34.20	2.38	34.20	1.95
Transport	13.50	46.27	13.50	57.34	13.50	73.58	13.50	43.34	13.50	49.88	13.50	66.52
Industry	5.40	11.44	5.40	19.28	5.40	33.68	5.40	20.27	5.40	19.28	5.40	33.68
Residential	7.20	25.22	7.20	33.39	7.20	25.57	7.20	23.66	7.20	27.78	7.20	25.58

gas flare reduction will be a good candidate for consideration in a cost-minimizing algorithm without any additional constraint such as CO₂ emission constraints. The issue of gas flare reduction has been a controversial one over the years which has finally resulted in the Nigerian government directive to all oil exploring companies to make year 2004 as the deadline for zero gas flare as revealed by the Minister of State for Environment (The Guardian 2000). This study has therefore factored this government policy in the model using cost and other associated data from the World Bank (1993) report on a planned gas flare reduction project. The most important target for CO₂ reduction in the Nigerian energy system is gas flare reduction (Adegbulugbe et al. 1996). Under all scenario assumptions, the quantity of gas being flared in 2030 with the 35% CO₂ reduction strategy will be comparable with the 1990 average for OECD countries (CEDEGAS 1993).

Under the 20% CO₂ reduction strategy, emissions in Nigeria in 2030 will be 1.77, 2.26 and 2.90 times the base year value for the LEG, MEG and HEG scenarios respectively (Table XIV). At 35% CO₂ emission reduction strategy, CO₂ emissions in 2030 will be 1.4, 1.8 and 2.38 times the base year value in the LEG, MEG

TABLE XVI
CO₂ emission ranking of renewable energy technologies (Ibitoye and Akinbami 1999)

Renewable energy technology	Incremental cost US\$	CO ₂ reduction potential million ton	Cost per unit of CO ₂ reduced \$/Mg of CO ₂
Small scale hydro power plant (<10 MW)	-427	41.31	-10.34
Large scale hydro power plant	-686	197.35	-3.48
Central solar PV plant	-24	18.74	-1.28
Residential solar PV plant	74	5.88	12.58

and HEG scenarios respectively. Carbon dioxide emissions from major sectors of the energy system under 20% and 35% CO₂ emissions reduction strategies are as shown in Table XV. The largest reduction is in the amount of gas flared followed by emission reductions in the electricity generation and the transport sector.

5.6. RENEWABLE ENERGY AND THE ENVIRONMENT

Renewable energy resources are useful in tackling the issues of global warming. A wide concern is the sustainability of existing development models and the role of renewable energy in reshaping the global energy trends (McGowan 1991). Carbon-dioxide reduction capacities of the renewable technologies used in this study were projected based on their maximum achievable market penetrations from possible future economic and social development in Nigeria (Ibitoye et al. 1998a). The ranking of the CO₂ mitigation potential of the renewable energy technologies used in this study was carried out on the basis of incremental cost per unit of CO₂ reduced in the system. The ranking of mitigation technologies reveals the small-scale hydro power technology (<10 MW) to be the most viable renewable energy technology for mitigating CO₂ emissions in the Nigerian electric power system at a negative incremental cost of \$10.34/Mg of CO₂ reduced, followed by large-scale hydro power technology, central solar photovoltaic system and residential solar photovoltaic technology. The incremental cost per unit of CO₂ reduced in the system for the other technologies shows that while the small-scale hydro power, large-scale hydro power, and the central solar PV plants are cost-effective, no-regrets CO₂ mitigation technologies, the residential solar PV plant is an expensive CO₂ mitigation technology in the Nigerian energy system. Large hydro power technology is the only renewable energy technology for electricity production that is favoured under both non-CO₂ and CO₂ reduction strategies. The other renewable energy

technologies examined enter into electricity supply mix only under severe CO₂ emission reduction scenarios.

6. Barriers to the Development and Penetration of Renewable Energy Technologies in Nigeria

Renewable energy resources other than hydro (particularly large-scale) and traditional biomass are currently not given any consideration in the national energy supply mix and can even only account for a tiny contribution in the decades to come. Thus, Nigeria may fall into a fossil fuel trap inspite of the available renewable energy resources. Nigeria has a lot to gain by promoting the use of renewable energy resources and technologies (Akinbami 1997). First, the bulk of the nation's populace is still in the rural areas which need energy security. Nigeria also has a diverse geography which may make it difficult to provide electricity to rural dwellers through the national grid. Renewable energy resources should offer good strategies for effective energy decentralization and security to both rural and urban dwellers. Also, through the use of renewable energy resources and technologies in the country, some ecological problems such as deforestation, desertification, greenhouse gases emissions, as well as soil erosion could be curtailed. Some of these challenges to the development, application and dissemination of renewable energy resources and technologies in Nigeria are discussed in this section.

6.1. TECHNOLOGICAL INCAPABILITY

Apart from solar thermal and biogas technologies, no other renewable energy technology has been developed in the country. Hence, these technologies have to be imported, thereby escalating further the already high investment costs of these technologies. Although, renewable energy technologies are and will remain of crucial importance in developing countries, their development is likely to be based in the industrialized world for the foreseeable future. This is because only industrialized countries have the technological base, the capital, and the infrastructure required to push large-scale new developments in the energy sector (Grubb 1990). For example, most of the solar PV system failures that occurred in Oman were due to harsh climatic conditions which were not taken into consideration while designing a PV system and selecting components (Myles 1992). Hence, an environment for effective renewable energy technology transfer from the developed into the developing countries has to be put in place as well as the enabling environment for the developing countries to acquire and adequately adapt such to suit local conditions. The provision of the UNFCCC, mandating the Annex I Parties (the developed countries) to transfer environmentally sound technology to developing countries, has not been implemented to any significant degree.

6.2. HIGH COST OF ENERGY INFRASTRUCTURE

This study revealed that under a least-cost optimization of Nigeria's energy system and under both non-CO₂ and CO₂ emissions constraints, large hydro power technology is the only preferred renewable energy technology. Economies of scale will result in lower investment cost per unit of installed capacity. Small scale hydro power, central and residential solar PV technologies have not penetrated Nigeria's energy supply system because of their relatively high investment costs (Table V). Another study (Akinbami et al. 2001) examined the cost-benefit analysis of a family-sized biogas digesters. The issue of high initial investment cost was also found to be one of the major barriers to popularizing the technology in Nigeria. Manufacturing and other costs of renewable energy technologies are declining and may still fall further as a result of improved design and manufacturing processes and growing markets. The notion that renewable energy sources are too expensive to be used substantially in the short or medium term is generally based on a very narrow definition of energy costs globally. For instance, cost analysis and comparisons often consider only the internal cost elements involved in the production and distribution of various energy forms. The external costs to society derived from burning fossil fuels or from nuclear energy generation are not included. The costs are external to the market process and the economic agents producing them, whereas renewable energy sources, in comparison to the conventional energy sources, produce very few globally impacting negative external effects, hence less external costs. This discrepancy which is not reflected in energy costs causes a serious distortion of energy costs in the market-oriented global energy market (Hohmeyer 1992). This market imperfection has to be phased out in the global energy market in order to promote renewable energy in Nigeria and other developing countries in accordance with the Kyoto Protocol Article 2(v) which requires for the Annex I Parties 'progressive reduction or phasing out of market imperfections, fiscal incentives, tax and duty exemptions and subsidies in all GHG emitting sectors that run counter to the objective of the Convention and apply market instruments.' Hence, there has to be a new energy supply technologies cost strategy that takes the existing distortion into consideration in the global energy market (Renewable Energy World 1999).

6.3. FINANCIAL CONSTRAINT

Considering that the energy sector accounts for less than 15% of national GDP and that Nigeria's annual budget is less than \$6 billion, coupled with the relatively high unit cost of installed capacity of renewable energy technologies, it becomes more difficult to publicly finance renewable energy in the country. The situation is even more serious in view of the enormous debt burden under which the Nigerian economy still suffers. Cumulative total energy system cost has been estimated to be about \$38 billion between 1990–2030 under the baseline strategy of the LEG scenario, while it would be about \$56 and \$98 billion under the MEG and

TABLE XVII
Cumulative total energy system costs (Adegbulugbe et al. 1996)

	LEG scenario			MEG scenario			HEG scenario		
	Total system cost (\$m)	Additional cost (\$m)	% Change	Total system cost (\$m)	Additional cost (\$m)	% Change	Total system cost (\$m)	Additional cost (\$m)	% Change
Baseline	38188	–	–	56029	–	–	97771	–	–
20% CO ₂ reduction	37205	–983	–2.57	54048	–1981	–3.54	94672	–3099	–3.17
35% CO ₂ reduction	39126	938	2.46	59197	3168	5.65	106332	8561	8.76

TABLE XVIII
CO₂ mitigation technologies evaluated in this study (Ibitoye and Akinbami 1999)

Mitigation technology	Incremental cost US\$	CO ₂ reduction capacity million ton	Cost per unit of CO ₂ reduced \$/Mg
CFL lighting	–299	5.155	–58.00
Improved kerosene stove	–131	6.122	–21.40
Displacement of fuel-oil by gas in cement industry	–138	7.49	–18.42
Improved electrical appliances in the residential sector	–161	9.566	–16.83
Efficient motors in industry	–171	10.738	–15.92
Small-scale hydro (<10 MW)	–427	41.313	–10.34
Kainji hydro power plant (Retrofit)	–351	50.01	–7.02
Improved woodstove in residential sector	–72	18.369	–3.92
Large-scale hydro	–686	197.353	–3.48
Central solar	–24	18.735	–1.28
Improved refrigerators	154	15.793	9.75
Residential solar PV	74	5.883	12.58
Gas flare reduction	45534	919.201	49.54
Efficient gasoline cars	17478	247.05	70.75
Improved air conditioners	218	1.54	141.56
Efficient diesel trucks	9060	60.096	150.76
Improved electrical appliances in the industrial and commercial sectors	2485	14.431	172.20

HEG scenarios respectively (Adegbulugbe et al. 1996). The total energy system cost includes the capital costs, the net expenditure on fuel, and other expenditure such as operating and maintenance costs, fuel delivery costs, etc. Even though the cumulative additional total system costs reveal that the 20% CO₂ emission reduction strategy can be implemented at a negative cost, the cumulative total energy system cost is still very high (Table XVII). A CO₂ mitigation assessment of the energy sector indicates some viable GHG mitigation options, some of which can be implemented at negative costs to the total energy system cost (Table XVIII). In carrying out the 20% CO₂ emission mitigation strategy and at a minimum cost to the economy, the model made use of more cost effective technologies across all the sectors of the economy. This reveals that it is more cost effective and beneficial to adopt the energy development path given by the 20% CO₂ reduction strategy under any scenario. Because, not only will CO₂ emissions be curtailed, money will also be saved. However, even if Nigeria were to adopt this development path, public funding cannot bear their burden. Hence, innovative financing techniques for the national energy system and particularly for both installation and maintenance of renewable energy facilities are needed to ensure adequate funding.

6.4. LOW LEVEL OF PUBLIC AWARENESS

The public awareness of renewable energy sources and technologies in Nigeria and their benefits both economically and environmentally are generally low. Consequently, the public is not equipped to influence the government to begin to take more decisive initiatives in enhancing the development, application, dissemination and diffusion of renewable energy resources and technologies in the national energy market. This may be due to the general Nigerian public illusion that the country is blessed with unlimited fossil fuels (oil, natural gas, coal). In spite of these abundant fossil fuels in the country, unreliable and inadequate supply of electricity and petroleum products are causing energy insecurity in the country. Hence, dissemination of decentralized renewable electric energy technologies will help raise the supply reliability of electricity and thereby reduce energy insecurity, offset fossil fuelled grid electricity and reduce carbon dioxide emissions in the country.

7. Policy Recommendations

7.1. A COMPREHENSIVE NATIONAL ENERGY POLICY

In the 40 years existence of Nigeria as a sovereign country, it has not had a comprehensive national energy policy. The nation can only boast of sub-sectoral energy policies. A comprehensive national energy policy will help the country to develop an integrated sustainable energy strategy which is pivotal to using energy

more efficiently and development and disseminate renewable energy technologies. A comprehensive policy will also act as a catalyst in the drive for transfer and acquisition of appropriate technologies adopted to suit local conditions and environment.

7.2. TECHNOLOGY TRANSFER AND ACQUISITION

Both the UNFCCC and the Kyoto Protocol require that all Parties take steps that are consistent with mitigation of climate change. The UNFCCC Article 4 Paragraph 5 requires that the Annex I Parties take steps to transfer environmentally sound technology to developing countries. Similarly, Paragraph (c) of Article 10 in the Kyoto Protocol requires that all Parties 'cooperate in the promotion of effective modalities for the development, application and distribution of, and take all practicable steps to promote, facilitate and finance, as appropriate, the transfer of, or access to, environmentally sound technologies, know-how, practices and processes pertinent to climate change, in particular to developing countries, including the formulation of policies and programmes for the effective transfer of environmentally sound technologies that are publicly owned or in the public domain and the creation of an enabling environment for the private sector, to promote and enhance the transfer of, and access to, environmentally sound technologies'. The Clean Development Mechanism (CDM), a mechanism financed by industrialized countries, aims to promote sustainable development in developing and newly industrialized countries. UN FCCC Kyoto Protocol Article 12 states that 'emission reductions only be certified when projects produce 'real, measurable, and long term benefits related to the mitigation of climate change' and 'reductions in emissions that are additional to any that would occur in the absence of the certified project activity' (Renewable Energy World 1999). While renewable energy systems are ideal candidates for the large-scale technology transfer the provisions of the UNFCCC and the Kyoto Protocol have not been implemented to any significant degree. There is a large potential for CO₂ emission reduction in Nigeria which may qualify for the CDM. However, workable guidelines for the CDM implementation have to be fashioned by both the developing and the developed countries. For Nigeria and other developing countries to realize the benefits of renewable energy system technology transfer, strong institutions must be established. This will help in adapting the technologies to the local environment as well as enhance the renewable energy systems learning curve.

7.3. INSTITUTIONAL SUPPORT AND STRENGTHENING

The development, application and diffusion of renewable energy systems especially for decentralized electricity generation and supply into the Nigerian economy require adequate institutional support and strengthening. Presently, there are two national energy research centres devoted to the science and technology of solar energy. The Energy Commission of Nigeria, a supervising clearing house, may

be insufficient for effective implementation of programmes for the development, application and diffusion of renewable energy systems. The establishment of a national agency for renewable energy systems may be needed. The Agency objectives should include:

- quantitative assessment of all renewable energy resources in the country;
- quantitative assessment of locally available manpower on various renewable energy systems;
- capacity building through training at various levels on the science and technology of renewable energy systems;
- development, acquisition and dissemination of appropriate renewable energy technologies;
- development of a programme of research, development and demonstration for renewable energy systems;
- financial support for the development, application and diffusion of renewable energy systems;
- outreach programmes to raise the level of awareness of the potentials and benefits of renewable energy systems to both the decision makers and the general public.

Acknowledgement

The author gratefully acknowledges the comments and suggestions of Professor O.O. Ogunkoya in the further review of this paper.

References

- Adegbulugbe, A.O., Ibitoye, F.I. and Akinbami, J-F.K.: 1996, *Greenhouse Gases Emission Reduction in Nigeria – Least-cost Reduction Strategies and Macroeconomic Impacts, Volume 3a: Energy Sector*. A Study Report presented to US Country Studies Program, Washington, D.C.
- Akinbami, J-F.K., Ilori, M.O., Oyebisi, T.O., Akinwumi, I.O. and Adeoti, O.: 2001, 'Biogas energy use in Nigeria: Current status, future prospects and policy implications', *Renewable and Sustainable Energy Review*, 5, 97–112.
- Akinbami, J-F.K.: 1997, 'Comparative environmental effects and cost analysis between conventional and non-conventional energy source – A case for objective analysis and decision making in Nigeria's energy policy', *Nigerian J. Renewable Energy* 5(1&2), 131–139.
- Annual Energy Outlook: 1992, Table C6.
- BEICIP, NNPC/NGC: 1991, *Nigeria Gas Utilization Study*, Interim Report DIP-90103, Volume 3.
- BTM Consult ApS: 1999, *International Wind Energy Development: World Market Update 1998*, Denmark, BTM Consult.
- CEDEGAS: 1993, *Natural Gas Statistics*, Paris.
- Electric Power Research Institute (EPRI): 1993, EPRI TAG Technology, #22.1C.
- Environmental Protection Agency: 1995, EPA's Inventory of Technologies, p. 74.
- Federal Office of Statistics (FOS): 1980, 1985, 1990, 1995, *Annual Abstract of Statistics*, Nigeria.
- Federal Office of Statistics (FOS): 1992, *Preliminary Report of General Household Survey*, Nigeria.

- Fishbone, L. G., Giesen, G., Goldstein, G., Hymmen, H.A., Stocks, K.J., Vos, H., Wilde, D., Zolcher, R., Balzer, C. and Abilock, H.: 1983, *User's Guide for MARKAL*, BNL-51701, Upton, NY, Brookhaven National Laboratory.
- Fujii, Y. and Yamaji, K.: 1998, 'Assessment of technological options in the global energy system for limiting the atmospheric CO₂ concentration', *Env. Econ. and Policy Stud.* **1**, 113–139.
- Grubb, M.J.: 1990, 'The Cinderella options: a study of modernized renewable energy technologies. Part 1 – A technical assessment', *Energy Policy* **18**(6), 525–542.
- Grubb, M.J. and Meyer, N.I.: 1993, 'Wind energy resources, systems and regional strategies', in T.B. Johansson, H. Kelly, A.K.N. Reddy, and R.H. Williams (eds.), *Renewable Energy: Sources for Fuels and Electricity*, Washington DC, Island Press.
- Hohmeyer, O.: 1992, 'Renewables and the full costs of energy', *Energy Policy* **April**, 365–375.
- Holdren, J.P., Morris, G. and Mintzer, I.: 1980, 'Environmental aspects of renewable energy sources', *Ann. Rev. Energy* **5**, 241–291.
- Ibitoye, F.I., Adegbulugbe, A.O. and Akinbami, J-F.K.: 1998, 'The cost of carbon-dioxide abatement in Nigeria's energy sector', *Int. J. of Global Energy Issues* **10**(2/4), 133–144.
- Ibitoye, F.I., Akinbami, J-F.K. and Adegbulugbe, A.O.: 1998, 'Optimal structure of Nigeria's energy sector under carbon-dioxide reduction', *Int. J. of Global Energy Issues* **10**(2/4), 145–154.
- Ibitoye, F.I. and Akinbami, J-F.K.: 1999, 'Strategies for implementation of CO₂ mitigation options in Nigeria's energy sector', *Applied Energy* **63**(1), 1–16.
- IIASA/WEC: 1998, *Global Energy Perspective*, in N. Nakicenovic, A. Grubler and A. McDonald (eds.), International Institute for Applied Systems Analysis (IIASA) and World Energy Council (WEC), Cambridge University Press, Cambridge, 300 pp.
- IPCC" 1995, *Guideline for National Greenhouse Gas Inventories*. Volume 3.
- Larson, E.D., Williams, R.H. and Johansson, T.B.: 1999, *Future Demands on Forests as a Source of Energy*, New York, USA, Centre for Energy and Environmental Studies, Princeton University, United Nations Development Program.
- Lovejoy, D.: 1984, *Mission Report on Nigeria: Programme on Demonstration of Solar and Wind Energy in Northern Nigeria*, New York: United Nations Department of Technical Cooperation and Development/United Nations Development Programme.
- McGowan, F.: 1991, 'Controlling the greenhouse effect: The role of renewables', *Energy Policy* **19**(2), 110–118.
- Munasinghe, M. 1999, 'Development, equity and sustainability (DES) in the context of climate change,' in Mohan Munasinghe and Rod Swart (eds.), *Climate Change and its Linkages with Development, Equity and Sustainability*, Proceedings of the IPCC Expert Meeting held in Colombo, Sri Lanka, 27–29 April 1999.
- Myles, A.S.: 1992. 'Solar PV system design parameters for Oman: A case study', in A.A.M. Sayigh (ed.), *Renewable Energy: Technology and the Environment*, Proceedings of the 2nd World Renewable Energy Congress, Reading, U.K. **1**, pp. 250–254. Published by Pergamon Press Ltd.
- Nakicenovic, N., Victor, N. and Morita, T.: 1998, 'Emissions scenarios database and review of scenarios', *Miti. Adapt. Strat. Global Change* **3**, 95–120.
- Ojosu, J.O. and Salawu, R.I.: 1990, 'Wind energy development in Nigeria', *Nig. J. Solar Energy* **9**, 29–32.
- Onyebuchi, E.I.: 1989, 'Alternative energy strategies for the developing world's domestic use: A case study of Nigerian household's final use patterns and preferences', *The Energy Journal* **10**(3), 121–138.
- Rave, K.: 1999, 'Who? What? Why? Wind power and the finance industry', *Renewable Energy World* **2**(5), 22–31.
- REW 1999, 'Wind force 10: How wind can produce 10% of world power by 2020', *Renewable Energy World* **2**(6), 40–61.
- Saboohi, Y.: 1989, *Model for Analysis of Demand for Energy (MADE-II)*, Stuttgart, IKE 8-19.

- Shell: 1996, *The Evolution of the World's Energy Systems*, London, UK, Shell International Ltd., Group External Affairs, 6 pp.
- Stefansson, V. and Fridleifsson, I.B.: 1998, 'Geothermal energy – European and worldwide perspective', Presented at Expert hearing on *Assessment and Prospects for Geothermal Energy in Europe*, Sub-Committee on Technology Policy and Energy of the Parliamentary Assembly of the Council of Europe, Strasbourg, France, pp. 10.
- Swisher, J., Wilson, D. and Schrattenholzer, L.: 1993, 'Long term strategies for mitigating global warming: Renewable energy potential,' *Energy* **8**(5), 437–459.
- The Guardian: 2000, *Govt. accepts new gas flaring proposal*, The Guardian Newspaper, Wednesday, August 16, pp. 23&25, Lagos.
- UNEP Collaborating Centre on Energy & Environment: 1994, 'UNEP Greenhouse Gas Abatement Costing Studies', Phase Two, Appendix: Guidelines. Riso, May.
- United Nations: 1981, *Report of the United Nations Conference on New and Renewable Sources of Energy*, A/CONF.100/11, New York: United Nations.
- UNCNRSE: 1981, 'Conference news: UNDP study foresees wider use of NRSE', *ASSET* **3**, 5.
- WCED (World Commission on Environment and Development): 1987, *Our Common Future (The Brundtland Report)*, Oxford, Oxford University Press, p. 195.
- WEC: 1994, *New Renewable Energy Resources: a Guide to the Future*, London, World Energy Council, published by Kogan Page Ltd., 391 pp.
- World Atlas: 1998, *1998 World Atlas & Industry Guide*, Sutton, UK, The International Journal on Hydropower & Dams, Aqua-Media International.
- World Bank: 1983, *Nigeria: Issues and Options in the Energy Sector*, Report No. 4440-UNI.
- World Bank: 1993, *Nigeria: Escravos Flared Gas Reduction Project*, Report No. 12178-UNI.
- World Bank: 1993a, *Nigeria: Issues and Options in the Energy Sector*, Report No. 11672-UNI.

